

Foundation (Jalapeno)

- Q1.** Astronomers recorded 2 · 5 planets with atmospheres and 3 · 2 without. What is the total number of planets recorded?
(A) 16
(B) 12
(C) 10
(D) 14
(E) 13
- Q2.** A space lab experiment had 15 successful launches and 8 failed launches. What is the relative frequency of successful launches?
(A) $\frac{15}{23}$
(B) $\frac{8}{15}$
(C) $\frac{8}{23}$
(D) $\frac{15}{8}$
(E) $\frac{23}{15}$
- Q3.** Which variable is most likely associated with the number of planets with rings in our solar system?
(A) Distance from the Sun
(B) Planet size
(C) Atmospheric composition
(D) Number of moons
(E) Temperature range
- Q4.** A two-way table shows 12 planets with high gravity and 7 planets with low gravity. If 5 planets have both high gravity and many moons, how many planets have many moons in total?
(A) 14
(B) 12
(C) 10
(D) 7
(E) 19
- Q5.** What is the purpose of using a two-way table in analyzing planetary data?
(A) To compare means
(B) To calculate medians
(C) To identify correlations between variables
(D) To determine the mode
(E) To find the range
- Q6.** A table shows 18 planets with solid surfaces and 12 planets with gas surfaces. What is the total number of planets if 8 have both solid and gas surfaces?
(A) 22
(B) 20
(C) 18
(D) 14
(E) 10
- Q7.** If a two-way table has x rows and y columns, what is the total number of cells in the table?
(A) $x + y$
(B) $x \cdot y$
(C) $x - y$
(D) x/y
(E) $y - x$
- Q8.** Astronomers collected data on 25 planets. The two-way table shows 10 planets with high temperatures and 8 planets with low temperatures. What is the number of planets with medium temperatures?
(A) 7
(B) 5
(C) 10
(D) 12
(E) 15

Open Ended Questions (FRQ)

- Q9.** Create a two-way table to analyze the relationship between the number of moons and the size of planets in our solar system. Use the following data: 5 small planets with many moons, 3 large planets with few moons, 2 small planets with few moons, and 1 large planet with many moons.
- Q10.** A space agency conducted an experiment with 20 spacecraft. The data showed 12 successful landings on planets with solid surfaces and 5 successful landings on planets with gas surfaces. Calculate the relative frequency of successful landings on planets with solid surfaces and interpret the results.

Chef's Tips

- Tip 1: Use visual aids to help students understand two-way tables.
- Tip 2: Encourage students to ask questions and explore real-world applications.
- Tip 3: Provide feedback and guidance to help students improve their problem-solving skills.